Advanced EVA POE Foam Solution

IUS-4065

Ultra-Soft, Low-Shrink Elastomer

A solution for softness, rebound and dimensional stability in EVA and POE foam formulations.

35 C	0.92%	60%
Foamed-product har	Thermal shrinkage	Ball rebound



R&D CHALLENGE

Why Ultra-Soft Midsoles Are Prone to Secondary Shrinkage

Low-hardness POE improves softness and feel, but secondary heating can relax the foam structure and cause edge shrinkage, warping, dimensional deviation and lamination misalignment. The real development challenge is to reduce hardness while maintaining rebound and controlling dimensional change after heating and lamination.

Softness

Lower finished-product hardness for improved cushioning, conformity and underfoot comfort.

Processing Stability

Reduce secondary dimensional change after baking, hot pressing and lamination.

Production Yield

Reduce left-right variation, lamination misalignment and batch-to-batch appearance variation.

MATERIAL SELECTION TRADE-OFF

The Material-Selection Trade-Off: Softer Materials Can Shrink More Under Heat

To achieve exceptional softness and high rebound, midsole formulations often use TAFMER™ or ENGAGE™ POE. TAFMER™ DF640 and ENGAGE™ 8842 typically measure Shore A 54–56 and offer excellent softness and feel, but their melting transition is below 50° C. Lamination, baking or hot pressing above this range can relax the foam structure and increase the risk of secondary shrinkage, edge contraction and warping.

Melting peak is not the only factor determining finished-product shrinkage. EVA ratio, crosslinking system, expansion ratio and secondary-processing conditions all affect the final result. This page uses a comparative formulation test at 70° C for 40 minutes for practical validation.

Material	Raw-material hardness	DSC melting peak	Primary advantage	Formulation and processing considerations
TAFMER™ DF640	Approx. 54–56A	Below 50° C	Good softness and feel	Dimensional stability after reheating must be validated
ENGAGE™ 8842	Approx. 54–56A	Below 50° C	Good softness and elasticity	Thermal shrinkage may occur after lamination and baking
IUS-4065	Approx. 40A	Approx. 64° C	Lower hardness and good flexibility	Wider thermal-processing window; finished-product shrinkage must still be validated in the actual formulation

ADVANTAGES

Product Advantages

Lower Hardness

In the same 60/40 PHR comparative formulation, IUS-4065 foam reaches Asker C 35 for a softer step-in and cushioning feel.

Wider Secondary-Processing Window

A DSC melting peak of approximately 64° C helps reduce premature softening and structural relaxation during lamination, baking or hot pressing.

Significantly Lower Thermal Shrinkage

Thermal shrinkage is 0.92% after 40 minutes at 70° C, approximately 71–74% lower than the DF610 and ENGAGE 8842 comparative formulations.

Improved Elongation

Elongation at break reaches 252.36%, approximately 9–11% higher than the two comparative formulations, demonstrating good flexibility.

Rebound Maintained

Even with substantially lower hardness, ball rebound remains at 60%, balancing softness and elasticity.

Formulation and Production Support

Starting formulations, samples, physical-property testing and production validation can be provided according to target hardness, expansion ratio, density and dimensional-shrinkage requirements.

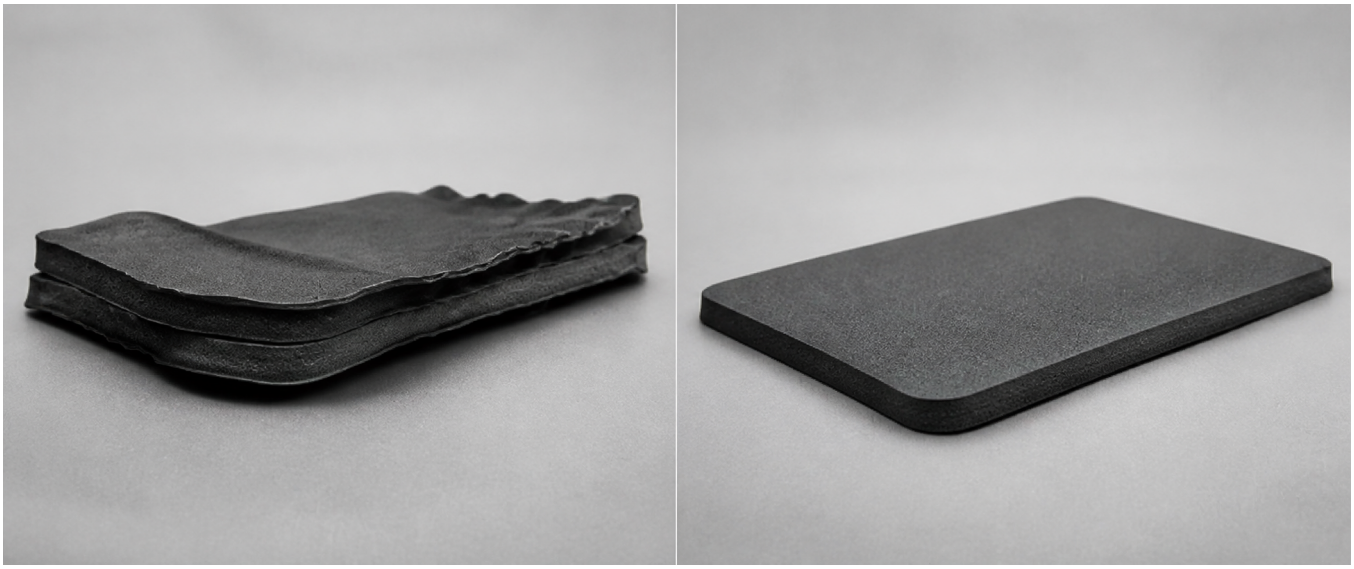
PRODUCT DESCRIPTION

Designed for EVA and POE Foam Systems

IUS-4065 is a high-performance elastomer modifier with a material hardness of approximately Shore A 40 and a DSC melting peak of approximately 64° C. A suitable EVA/POE ratio, crosslinking system and foaming

conditions can improve dimensional retention after secondary processing.

It is not a general-purpose TPE focused only on low hardness; it is designed to address the development challenge of balancing softness, rebound and dimensional stability.



Low-melting material heat deformation versus IUS-4065 dimensional stability

INTERNAL VALIDATION

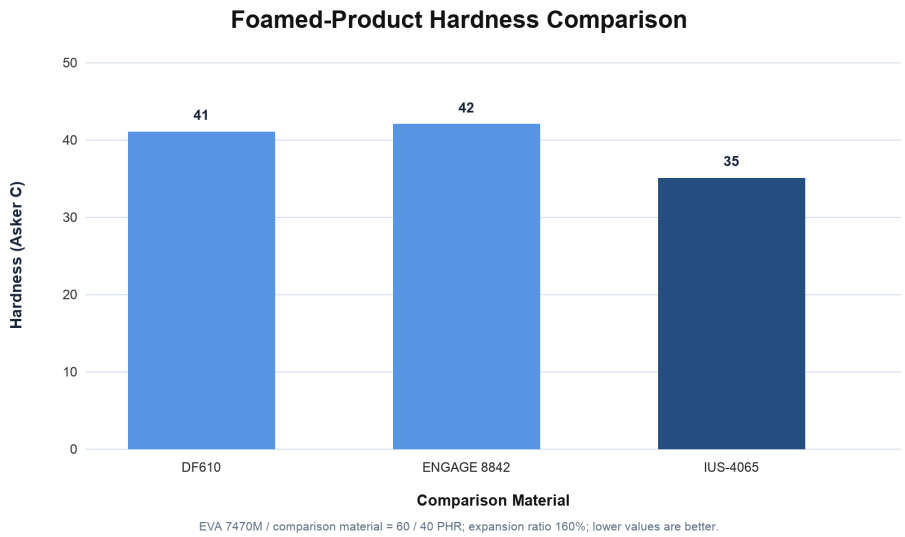
Internal Performance Validation

EVA 7470M/comparison material = 60/40 PHR; expansion ratio 160%. Data are typical values under the specified formulation and conditions.

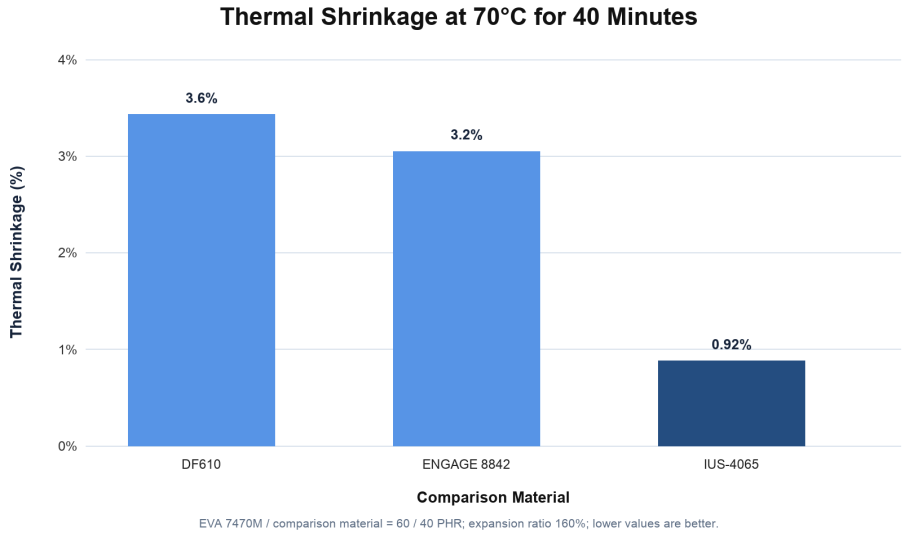
After testing at 70° C for 60 minutes, IUS-4065 shows only 1.38% thermal shrinkage and excellent heat-resistant dimensional stability. Request a sample to validate your formulation and processing conditions.

IUS-4065 provides lower hardness, higher elongation and substantially reduced thermal shrinkage. Tensile and tear strength are slightly lower and compression set is slightly higher, so the formulation should be optimized for actual product requirements.

Test item	Unit and condition	DF610	ENGAGE 8842	IUS-4065	Interpretation
Hardness	Asker C	41	42	35	Softer
Specific gravity	g/cm³	0.1826	0.1875	0.1816	Lowest
Tensile strength	kgf/cm²	19.73	19.05	18.25	Slightly lower but comparable
Elongation at break	%	227.65	231.43	252.36	Highest
Tear strength	kgf/cm	8.56	8.08	7.46	Optimize to requirements
Compression set	% 23° C 6h	28.50	29.12	31.67	Slightly higher
Thermal shrinkage	% 70° C 40min	3.60	3.20	0.92	Significantly lower
Ball rebound	%	60	61	60	Maintained
Expansion ratio	%	160	160	160	Same



IUS-4065 hardness is 35C, approximately 15–17% lower than the two comparative formulations.



Thermal shrinkage is 0.92%, approximately 71–74% lower than the two comparative formulations.

CHARACTERISTICS

Product Characteristics

Item	Typical IUS-4065 Data
Product type	Ultra-soft, low-shrink elastomer modifier
Material appearance	Translucent white pellets
Material hardness	Approx. Shore A 40
DSC melting peak	Approx. 64° C
Primary compatible systems	EVA/POE and elastomer foam formulations

Item	Typical IUS-4065 Data
Core function	Softening, rebound retention and improved dimensional stability after secondary processing
Applicable processes	Chemical foaming, injection foaming, compression molding, cold/hot pressing and lamination
Formulation approach	Adjust addition ratio according to target hardness, expansion ratio, mechanical properties and processing conditions

This information is for material selection and formulation-development reference only. Actual performance depends on formulation, processing and test conditions.